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5.1.1 CENTER OF MASS

When a rigid body moves, all particles that constitute the body need not take the same path. Depending on the type of motion, different particles of the body may take different paths. For example, when a wheel rolls on a surface, the path of the center point of the wheel and the paths of other points of the wheel are different. In this Unit, we study about the translation, rotation and the combination of these motions of rigid bodies in detail.

5.1.2 Center of Mass of a Rigid Body

When a bulk object (say a bat) is thrown at an angle in air as shown in Figure 5.1; do all the points of the body take a parabolic path? Actually, only one point takes the parabolic path and all the other points take different paths.

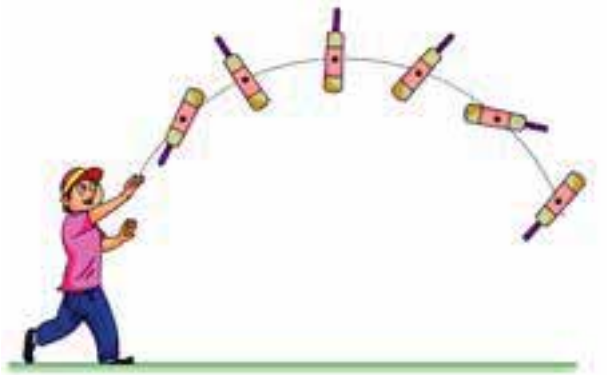


Figure 5.1 Center of mass tracing the path of a parabola

The one point that takes the parabolic path is a very special point called *center of mass* (CM) of the body. Its motion is like the motion of a single point that is thrown. *The center of mass of a body is defined as a point where the entire mass of the body appears to be concentrated.* Therefore, this point can represent the entire body.

For bodies of regular shape and uniform mass distribution, the center of mass is at the geometric center of the body. As examples, for a circle and sphere, the center of mass is at their centers; for square and rectangle, at the point their diagonals meet; for cube and cuboid, it is at the point where their body diagonals meet. For other bodies, the center of mass has to be determined using some methods. The center of mass could be well within the body and in some cases outside the body as well.

5.1.3 Center of Mass for Distributed Point Masses

A point mass is a hypothetical point particle which has nonzero mass and no size or shape. To find the center of mass for a collection of n point masses, say, m_1, m_2, m_3 . . . m_n we have to first choose an origin and an appropriate coordinate system as shown in Figure 5.2. Let, x_1, x_2, x_3 . . . x_n be the X-coordinates of the positions of these point masses in the X direction from the origin.

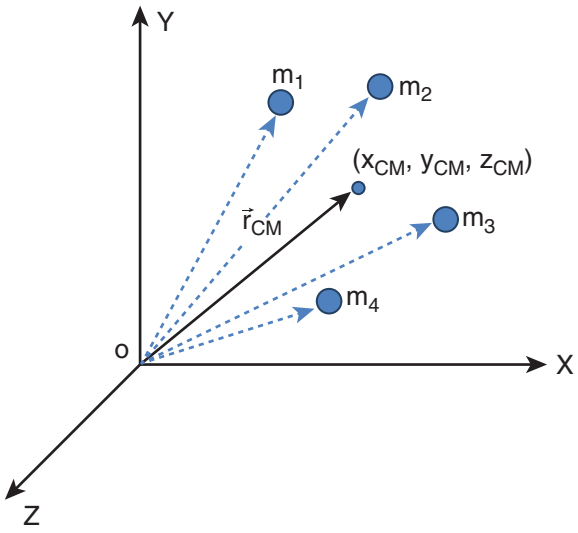


Figure 5.2 Center of mass for distributed point masses